Gamal Tawaf

SENIOR SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Senior Software Engineer and Google Cloud Professional Cloud Architect. 12+ years building high-reliability backends across payments and large-scale platforms - distributed systems, observability, and GCP, with Python as the primary language. Technical lead who grows engineering teams and compresses partner integrations from months to weeks.

SKILLS

Backend: Python, FastAPI, Flask, Django, asyncio, Java, Spring Boot, PHP, Laravel, REST API design

Frontend: React, Svelte, D3.js, JavaScript, HTML

Cloud & Architecture: Google Cloud Platform (Cloud Run, Pub/Sub, BigQuery, Vertex AI), Kubernetes, Docker, Terraform, microservices, distributed systems, high-availability design

Data & Real-Time: PostgreSQL, MySQL, Redis, BigQuery, Google Cloud Pub/Sub, real-time event pipelines

Observability & Delivery: OpenTelemetry distributed tracing, monitoring & alerting, GitHub Actions, GitLab CI/CD, pytest, JUnit, test automation

AI Engineering: Vertex AI, Gemini, LangChain, RAG systems, AI data pipelines

WORK EXPERIENCE

Senior Software Engineer

Truss Payments

Austin, Texas / Remote

Aug 2026 - Oct 2026

- Cut new-engineer local setup to under an hour by introducing a Docker-based development environment.
- Owned on-call response and resolved customer payment issues in production.
- Defined conventions for agent-written code so new agent work follows existing team patterns, language best practices, and coding conventions.
- Audited cloud infrastructure for compliance and documented the remediation path to close gaps.
- Reworked unfinished payment and account flows so OTP shows loading state and entered number, unavailable actions stay hidden, blocked steps notify the owner, and failures surface actionable errors.
- Designed and presented a GCP cost-reduction plan targeting ~50% savings via Committed Use Discounts and ~20% more via rightsizing and unused-capacity cleanup; leadership approved it for implementation.

Fractional CTO

Attendi

Remote

Apr 2026 - Present (fractional, concurrent with Fiserv and Truss)

- Hardened the platform for its first production customer: fixed security vulnerabilities, added audit trails for critical actions, and enforced least-privilege access.
- Cut hosting costs 50% by consolidating five services across two cloud providers into one GCP service and replacing manual provisioning with Terraform.
- Used AI-driven development so a lean team shipped backend, mobile, and UI work previously done by a multinational engineering team.
- Cut small-change delivery from weeks to under a day by wiring Jira tickets to pull requests that review and deploy on merge through CI/CD.
- Shipped mobile releases via over-the-air updates and Play Store / App Store pipelines.
- Delivered dashboard and mobile UX improvements alongside the faster delivery path.
- Added Google Cloud Monitoring alerts, golden-signal dashboards, and centralized logs, plus mobile crash analytics, for team and AI-assisted investigation.
- Deprioritized features that did not advance the company's primary objective.
- Onboarded the non-technical CEO onto the delivery workflow so he can request, ship, and verify production features end-to-end without waiting on engineering.

Senior Software Engineer

Fiserv (via Ascendion)

Toronto, ON

Jan 2026 - Aug 2026 (contract via Ascendion)

- Engineer on a core financial-services platform (Python, FastAPI, Flask, asyncio, MySQL, Redis, Pub/Sub, Kubernetes, Docker) at one of the world's largest fintech providers, serving as Feature Lead for multiple core platform features and owning cross-team delivery.
- Introduced OpenTelemetry distributed tracing with traceparent propagation across 24 microservices, cutting production issue diagnosis time by 80% - from several hours of cross-service investigation to under an hour.
- Re-architected vendor integration across 5 microservices into a configuration-driven design, cutting new-vendor onboarding from ~2 months of cross-service code changes to ~2 weeks of configuration.
- Led a new vendor integration initiative end-to-end, delivering a customer-facing capability designed to drive client retention and product adoption.
- Resolved on-call production incidents on a platform where reliability directly affects financial transactions.

Solution Architect / Engineering Team Lead

Inkeros

Etobicoke, ON

Jan 2025 - Jan 2026

- Recruited, led, and mentored the full engineering team (junior through senior) while owning architecture for the company's core product.
- Led discovery with internal clients in HR, Quality Assurance, and Sales: diagnosed each issue, broke it into a solution, and presented it for the team to build.
- Architected a mission-critical enterprise portal (Python/Django on GCP) automating sales-reporting workflows for 250+ staff - eliminating ~2 hours/day of manual submission per sales rep and hours of manager spreadsheet work, reducing the entire workflow to a 5-10 minute approval. Designed SaaS-ready for multi-tenant expansion.
- Built a fully automated CI/CD pipeline (Docker, GitHub Actions, Artifact Registry, Cloud Run) delivering one-click deployments at a 99.8% success rate.
- Designed and piloted an AI compliance system analyzing call-center audio with Gemini and Vertex AI - orchestrated via GCP Workflows into BigQuery - architected to scale to a 100-agent, 6-day/week calling operation for fraud prevention and compliance insight.
- Built generative-AI data pipelines (Pandas, GeoPandas, SERP/Maps APIs) that quadrupled lead-profiling throughput, and built and deployed a LangChain + Vertex AI RAG system.
- Ran the platform at 99.9% uptime on GCP autoscaling infrastructure (FastAPI and asyncio services, Kubernetes, Redis, Pub/Sub event pipelines) and cut mean time to resolution by 60% through improved monitoring and runbooks.

Senior Software Engineer

PERCS

Remote

Apr 2024 - Jan 2025 (layoff)

- Scaled full-stack platform features through rapid user growth and integrated product analytics to guide roadmap decisions.
- Designed and launched Web3 games on blockchain-based social protocols (Warpcast Frames) during the protocol's early growth stage, driving strong user engagement and raising the company's visibility in the ecosystem.

Senior Software Engineer

Binance, Card & Pay

Remote

Jul 2022 - Jul 2023 (layoff)

- Integrated 3 global payment partners' REST APIs to enable MasterCard issuance across Europe, the Middle East, and South America at one of the world's largest crypto exchanges, built with enterprise integration patterns.
- Migrated the card-ordering and KYC services from Laravel to Java/Spring Boot 3, cutting new payment-partner onboarding from ~3 months to ~1 week and improving performance and scalability of a revenue-critical flow.
- Built the new Spring Boot services with unit testing from day one (vs. an untested legacy codebase), and trained team members in Java and Spring Boot during the migration.

Senior Web Engineer

VerticalScope Inc.

Remote

Jul 2021 - Jun 2022 (recruited into Binance)

- Designed and shipped features for community platforms serving 10M+ users, and launched the platform's first marketplace product - a zero-to-one build converting ad-hoc forum-post listings into a structured commerce experience and new revenue stream.

Team Lead / Senior Software Developer

Afilias Canada

Toronto, ON

Feb 2016 - Jul 2021

- Led the in-house migration of a business-critical Ruby on Rails application off a third-party vendor - upgrading Rails v3 to v5 and migrating MySQL to PostgreSQL in 6 months - delivering $500K in annual savings and full operational control.
- Took ownership of 3 previously outsourced systems and built the internal team's capability to run them, establishing lasting business confidence in in-house engineering.
- Hired, trained, and led a team of 5-7 developers over 5+ years, growing engineers from junior to senior levels.
- Led discovery with the management team on business intelligence, then implemented a BI visualization tool (React, D3.js, Python backend connected to Qlik) so leaders could use data they already had.

CERTIFICATIONS

Google Cloud Professional Cloud Architect Google Cloud 2025-2027

Hugging Face AI Agents Course Hugging Face Jul 2026

EDUCATION

Bachelor of Computer Science
York University

Computer Programming Diploma
Seneca College